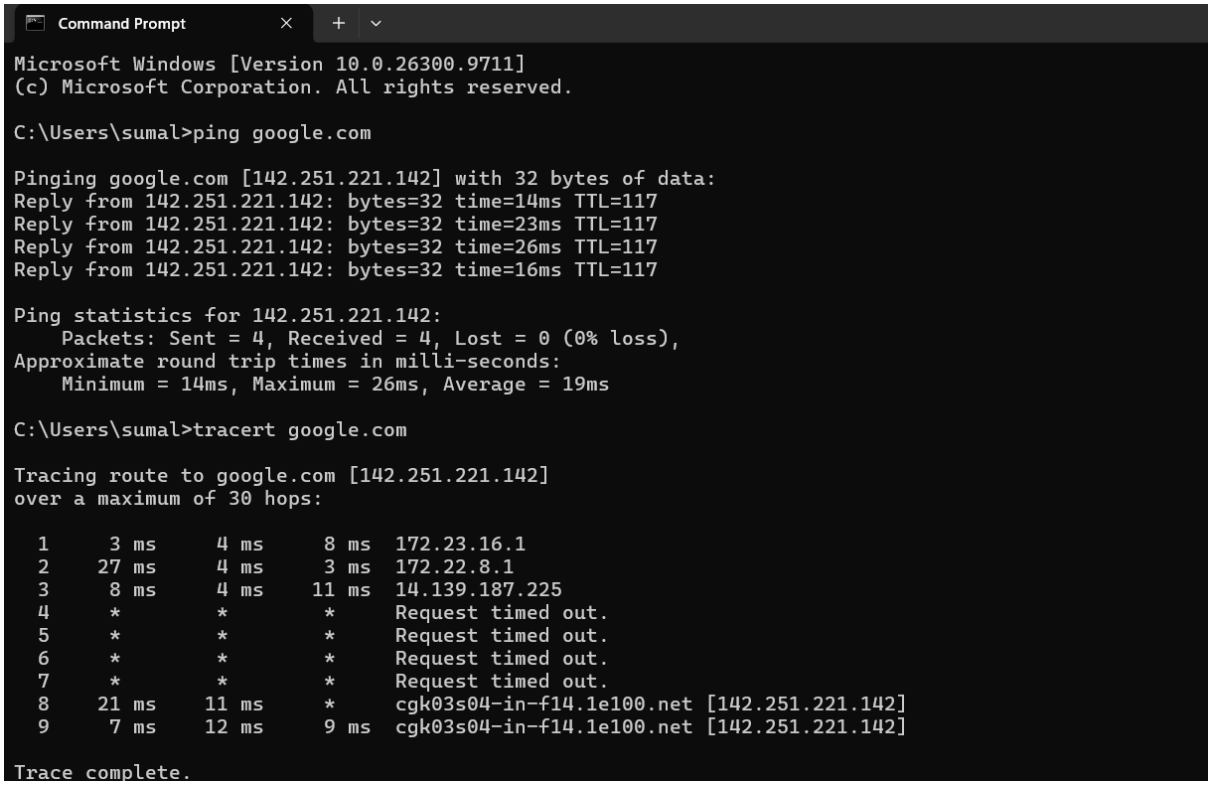
17. Windows OS Commands Execution (traceroute, ping, ifconfig, netstat)



```
Command Prompt

Microsoft Windows [Version 10.0.26300.9711]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sumal>ping google.com

Pinging google.com [142.251.221.142] with 32 bytes of data:
Reply from 142.251.221.142: bytes=32 time=14ms TTL=117
Reply from 142.251.221.142: bytes=32 time=23ms TTL=117
Reply from 142.251.221.142: bytes=32 time=26ms TTL=117
Reply from 142.251.221.142: bytes=32 time=16ms TTL=117

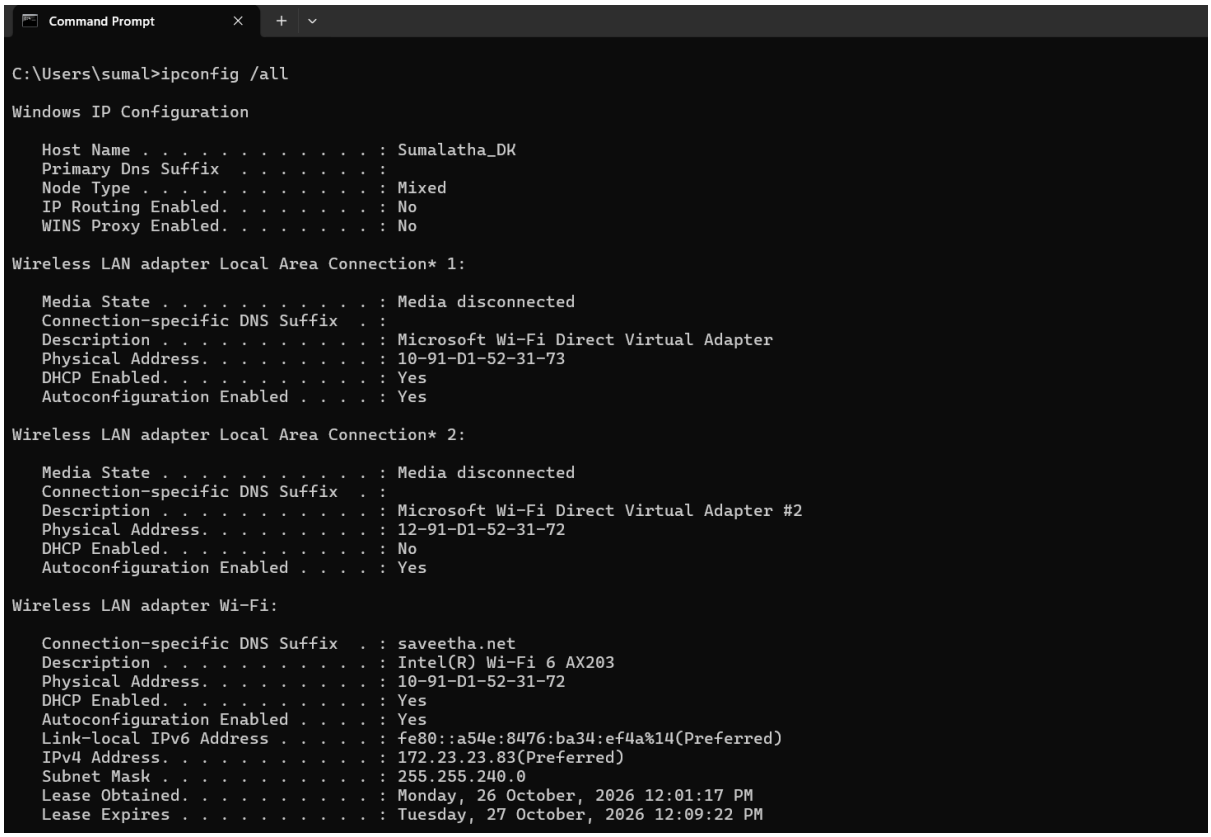
Ping statistics for 142.251.221.142:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 14ms, Maximum = 26ms, Average = 19ms

C:\Users\sumal>tracert google.com

Tracing route to google.com [142.251.221.142]
over a maximum of 30 hops:

  1     3 ms     4 ms     8 ms  172.23.16.1
  2    27 ms     4 ms     3 ms  172.22.8.1
  3     8 ms     4 ms    11 ms  14.139.187.225
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6     *        *        *     Request timed out.
  7     *        *        *     Request timed out.
  8    21 ms    11 ms     *     cgk03s04-in-f14.1e100.net [142.251.221.142]
  9     7 ms    12 ms     9 ms  cgk03s04-in-f14.1e100.net [142.251.221.142]

Trace complete.
```



```
Command Prompt

C:\Users\sumal>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : Sumalatha_DK
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Mixed
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Wireless LAN adapter Local Area Connection* 1:

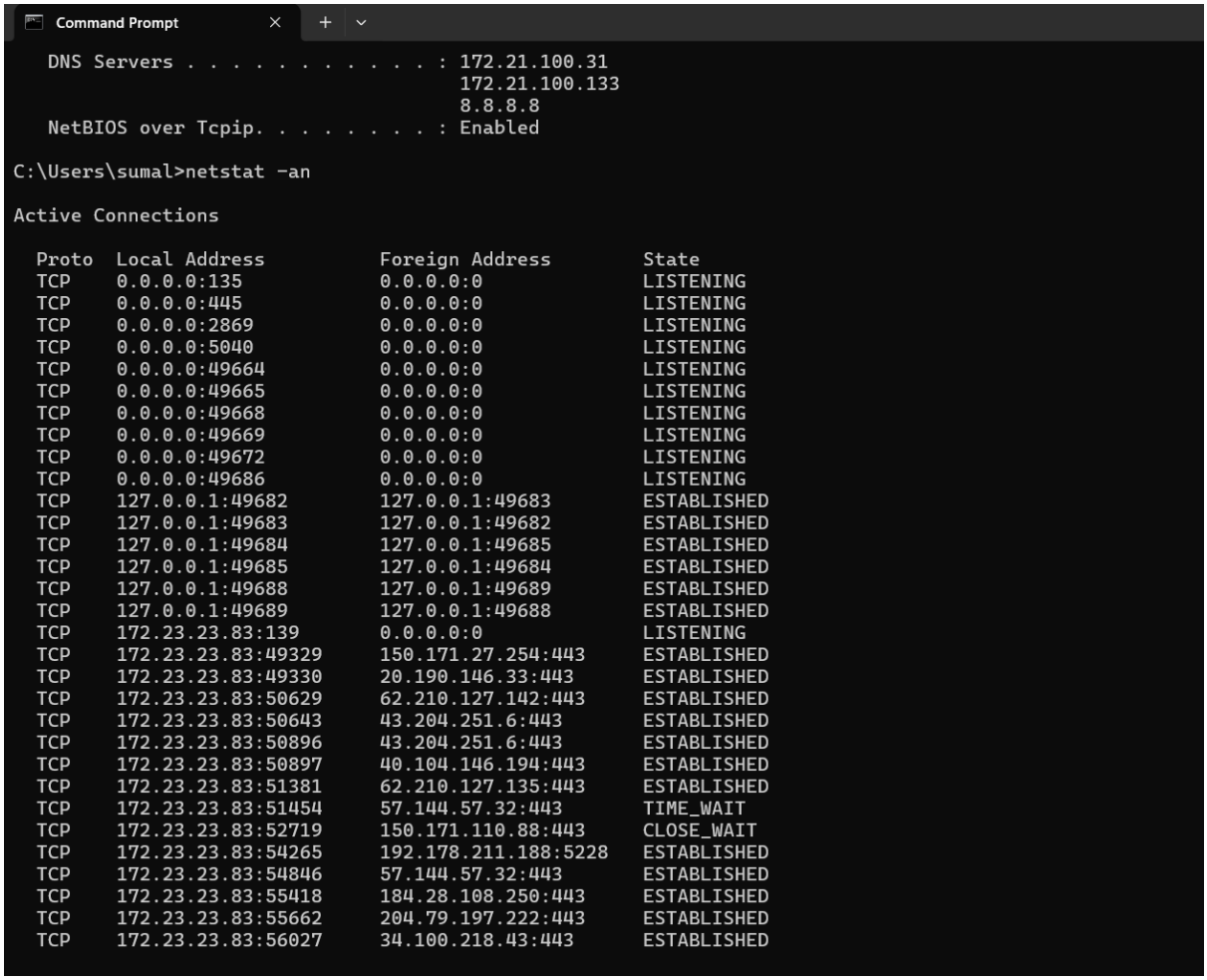
   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : 10-91-D1-52-31-73
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

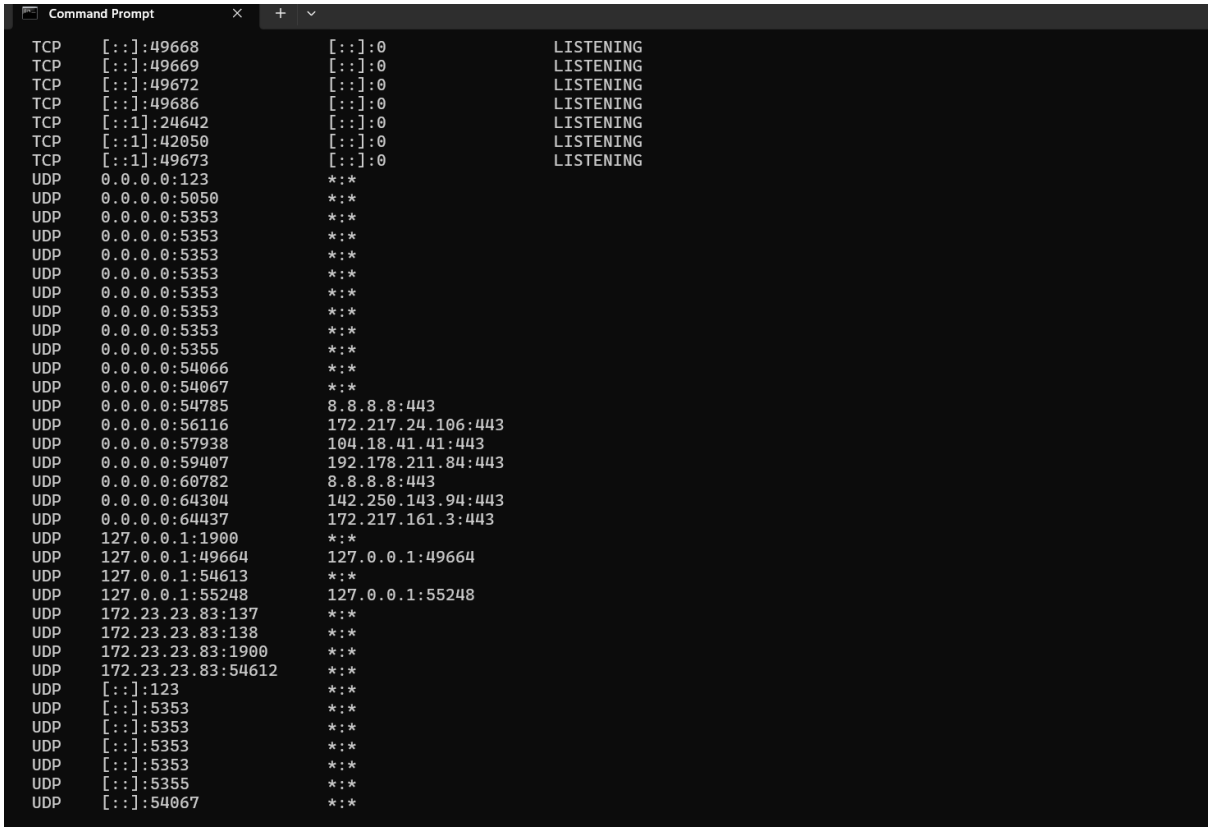
Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : 12-91-D1-52-31-72
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : saveetha.net
   Description . . . . . . . . . . . : Intel(R) Wi-Fi 6 AX203
   Physical Address. . . . . . . . . : 10-91-D1-52-31-72
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::a54e:8476:ba34:ef4a%14(Preferred)
   IPv4 Address. . . . . . . . . . . : 172.23.23.83(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Lease Obtained. . . . . . . . . . : Monday, 26 October, 2026 12:01:17 PM
   Lease Expires . . . . . . . . . . : Tuesday, 27 October, 2026 12:09:22 PM
```


Command Prompt
DNS Servers . . . . . : 172.21.100.31
172.21.100.133
8.8.8.8
NetBIOS over Tcpip. . . . . : Enabled
C:\Users\sumal>netstat -an
Active Connections
Proto Local Address Foreign Address State
TCP 0.0.0.0:135 0.0.0.0:0 LISTENING
TCP 0.0.0.0:445 0.0.0.0:0 LISTENING
TCP 0.0.0.0:2869 0.0.0.0:0 LISTENING
TCP 0.0.0.0:5040 0.0.0.0:0 LISTENING
TCP 0.0.0.0:49664 0.0.0.0:0 LISTENING
TCP 0.0.0.0:49665 0.0.0.0:0 LISTENING
TCP 0.0.0.0:49668 0.0.0.0:0 LISTENING
TCP 0.0.0.0:49669 0.0.0.0:0 LISTENING
TCP 0.0.0.0:49672 0.0.0.0:0 LISTENING
TCP 0.0.0.0:49686 0.0.0.0:0 LISTENING
TCP 127.0.0.1:49682 127.0.0.1:49683 ESTABLISHED
TCP 127.0.0.1:49683 127.0.0.1:49682 ESTABLISHED
TCP 127.0.0.1:49684 127.0.0.1:49685 ESTABLISHED
TCP 127.0.0.1:49685 127.0.0.1:49684 ESTABLISHED
TCP 127.0.0.1:49688 127.0.0.1:49689 ESTABLISHED
TCP 127.0.0.1:49689 127.0.0.1:49688 ESTABLISHED
TCP 172.23.23.83:139 0.0.0.0:0 LISTENING
TCP 172.23.23.83:49329 150.171.27.254:443 ESTABLISHED
TCP 172.23.23.83:49330 20.190.146.33:443 ESTABLISHED
TCP 172.23.23.83:50629 62.210.127.142:443 ESTABLISHED
TCP 172.23.23.83:50643 43.204.251.6:443 ESTABLISHED
TCP 172.23.23.83:50896 43.204.251.6:443 ESTABLISHED
TCP 172.23.23.83:50897 40.104.146.194:443 ESTABLISHED
TCP 172.23.23.83:51381 62.210.127.135:443 ESTABLISHED
TCP 172.23.23.83:51454 57.144.57.32:443 TIME_WAIT
TCP 172.23.23.83:52719 150.171.110.88:443 CLOSE_WAIT
TCP 172.23.23.83:54265 192.178.211.188:5228 ESTABLISHED
TCP 172.23.23.83:54846 57.144.57.32:443 ESTABLISHED
TCP 172.23.23.83:55418 184.28.108.250:443 ESTABLISHED
TCP 172.23.23.83:55662 204.79.197.222:443 ESTABLISHED
TCP 172.23.23.83:56027 34.100.218.43:443 ESTABLISHED


Command Prompt
TCP [::]:49668 [::]:0 LISTENING
TCP [::]:49669 [::]:0 LISTENING
TCP [::]:49672 [::]:0 LISTENING
TCP [::]:49686 [::]:0 LISTENING
TCP [::1]:24642 [::]:0 LISTENING
TCP [::1]:42050 [::]:0 LISTENING
TCP [::1]:49673 [::]:0 LISTENING
UDP 0.0.0.0:123 *:*
UDP 0.0.0.0:5050 *:*
UDP 0.0.0.0:5353 *:*
UDP 0.0.0.0:5353 *:*
UDP 0.0.0.0:5353 *:*
UDP 0.0.0.0:5353 *:*
UDP 0.0.0.0:5353 *:*
UDP 0.0.0.0:5353 *:*
UDP 0.0.0.0:5353 *:*
UDP 0.0.0.0:5355 *:*
UDP 0.0.0.0:54066 *:*
UDP 0.0.0.0:54067 *:*
UDP 0.0.0.0:54785 8.8.8.8:443
UDP 0.0.0.0:56116 172.217.24.106:443
UDP 0.0.0.0:57938 104.18.41.41:443
UDP 0.0.0.0:59407 192.178.211.84:443
UDP 0.0.0.0:60782 8.8.8.8:443
UDP 0.0.0.0:64304 142.250.143.94:443
UDP 0.0.0.0:64437 172.217.161.3:443
UDP 127.0.0.1:1900 *:*
UDP 127.0.0.1:49664 127.0.0.1:49664
UDP 127.0.0.1:54613 *:*
UDP 127.0.0.1:55248 127.0.0.1:55248
UDP 172.23.23.83:137 *:*
UDP 172.23.23.83:138 *:*
UDP 172.23.23.83:1900 *:*
UDP 172.23.23.83:54612 *:*
UDP [::]:123 *:*
UDP [::]:5353 *:*
UDP [::]:5353 *:*
UDP [::]:5353 *:*
UDP [::]:5353 *:*
UDP [::]:5355 *:*
UDP [::]:54067 *:*